

Assignment 2

Introduction to Machine Learning

CS-3410 / MAT-3211, Spring 2026

Date: 21st April, 2026

Instructions

- The assignment is worth **100 marks**, and there are 2 parts to it.
- Part A consists of theoretical problems, and you are expected to submit well organized LaTeX solutions on Google Classroom.
- Part B consists of coding problems, and you need to attach a link to a GitHub repository containing the **jupyter notebook(s)**. The date of submission considered will be the date of the **last commit** to the repository. All code must be in Python, and the only libraries allowed are **scipy, numpy, datasets, and matplotlib** unless specified otherwise.
- We encourage the use of LLMs and online resources provided that they are used to enhance your understanding and not to copy naively. You are required to disclose all sources you have used, which includes LLMs like ChatGPT, Gemini, etc.

Part A: Theoretical Problems

(55 marks)

Question 1

(10 marks)

Let $\mathcal{H}$ be a finite hypothesis class of binary classifiers with size $|\mathcal{H}| = k$. Assume all hypotheses are evaluated on a dataset of m independent and identically distributed (i.i.d.) examples. Let $\hat{\varepsilon}(h)$ denote the empirical training error and $\varepsilon(h)$ denote the true generalization error for a hypothesis $h \in \mathcal{H}$. Using Hoeffding's inequality and the union bound, prove that the sample complexity m required to guarantee that

$$P(\forall h \in \mathcal{H}, |\varepsilon(h) - \hat{\varepsilon}(h)| \leq \gamma) \geq 1 - \delta$$

is bounded by:

$$m \geq \frac{1}{2\gamma^2} \log \left(\frac{2k}{\delta} \right).$$

Question 2

(5 + 6 + 4 marks)

Suppose we have a centered dataset $X \in \mathbb{R}^{n \times d}$ (that is, the empirical mean of the rows is zero). Let

$$\Sigma = \frac{1}{n} X^\top X$$

be the sample covariance matrix. In PCA, we want to find a unit direction v (so $\|v\|_2 = 1$) that maximizes the variance of the projected data.

1. Show that maximizing the variance of the projected data is equivalent to maximizing

$$v^\top \Sigma v.$$

2. Write down the constrained optimization problem for maximizing

$$v^\top \Sigma v$$

subject to

$$v^\top v = 1.$$

Use a Lagrange multiplier and prove that the optimal direction v must be an eigenvector of Σ .

3. Consider the following 2D data:

$$(1, 1), (2, 2), (3, 2), (4, 3).$$

Do PCA on this dataset. You should:

- (a) find the direction of maximum variance mathematically,
- (b) compute the projections of the data onto this direction.

Then make the following plots:

- (a) the data points and the line along the direction of maximum variance;
- (b) the projected data points.

Question 3

(10 + 5 marks)

The Evidence Lower Bound (ELBO) is an objective used in latent-variable models, where we model observed data x using an additional unobserved variable z . It provides a tractable lower bound on the data log-likelihood, $\log p(x)$.

Read about the ELBO before answering. A good starting point is Matthew Bernstein's short note, *The evidence lower bound (ELBO)*: <https://mbernste.github.io/posts/elbo/>. You may also consult other resources.

1. Derive the ELBO as a lower bound on the data log-likelihood $\log p(x)$. Your derivation should clearly show:
 - (a) how an auxiliary distribution over the latent variable is introduced, and
 - (b) where Jensen's inequality is used.
2. Show how maximizing the ELBO is equivalent to minimizing an objective made up of:
 - (a) a **reconstruction term** (which measures how well the model explains the observed data), and
 - (b) a **KL-divergence regularization term** (which encourages the auxiliary distribution to stay close to the model's latent prior).

Write the result in mathematical form and briefly explain the role of each term.

Question 4**(5 marks)**

Consider an input sequence

$$s = w_1, w_2, \dots, w_n.$$

When this sequence is passed through a Transformer, it produces a sequence of output embeddings

$$O = [o_1, o_2, \dots, o_n].$$

Now, for any prefix of the input sequence,

$$s_i = w_1, w_2, \dots, w_i,$$

let the corresponding output sequence be

$$O_i = [o_1, o_2, \dots, o_i].$$

In a decoder-only Transformer architecture, prove that for every i ,

$$s_i = w_1 \cdots w_i \implies O_i = [o_1, \dots, o_i].$$

Explain why this property is important when training autoregressive language models.

Question 5**(2 + 5 + 3 marks)**

Imagine a server which can be in two states, idle and occupied (I, O). At any state I/O and time point $t = 0, 1, \dots$, you can choose to either stay in the same state or move to the other state. However, due to system errors, every choice you make ends up happening only with probability $1 - \epsilon$, with an $\epsilon \in (0, \frac{1}{2})$ chance that the opposite of your choice ends up happening. The system rewards occupied states with $+1$, and idle by -3 .

1. Model the above as an MDP, with discount factor $\gamma \in (0, 1)$.
2. Find the optimal policy and justify it.
3. Calculate the value function for this policy.

Part B: Coding Problems**(45 marks)****Question 1****(15 marks)**

Implement SVMs with soft margins by:

- (a) Taking the margin cost and data as input to generate an optimization problem.
- (b) Use `scipy` to solve the problem.
- (c) Find the support vectors, and use them to find the optimal w, b .

Then use the SVM you wrote on the breast cancer dataset. Report your findings.

Question 2**(15 marks)**

Implement the K -means algorithm from scratch. Use it on the Iris dataset for $k = 2, 3, 4, 5$, and clearly plot the clusters and the total discrepancy loss. Report your findings clearly.

Question 3**(15 marks)**

Use the Fashion-MNIST dataset and train an autoencoder on it. Use various different network sizes. Plot and report your findings on train and validation accuracy, along with time taken to train.